The Edge Negotiator

Trustworthy On-Device Small-Language-Model Control of Urban Traffic Signals with a Provable Accident Audit

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MSc Systems Engineering for the Internet of Things
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September 2026

This report is submitted as part requirement for the MSc in Systems Engineering for the Internet of Things at University College London. It is substantially the result of my own work except where explicitly indicated in the text. [TODO: replace with the exact departmental declaration wording]

Abstract

When a signalised junction is involved in a collision, someone must be able to prove what the controller knew and what it decided. Language-model traffic controllers are now competitive in simulation, but every published system is served from the cloud and none can answer that question. This dissertation asks whether a language model small enough to live in the signal cabinet can control real London junctions competently *and* bind every actuation into a record an investigator can trust — and what, if anything, fine-tuning contributes to that goal.

I build the full stack on one consumer device: a small language model served by Microsoft Foundry Local within a 6 GB envelope, proposing signal phases behind a MaxPressure shield and anti-starvation floor, with every decision Ed25519-signed into a per-junction hash chain and reconstructable after a collision into a cited UK-law evidence pack. I distil a frontier teacher into 0.6B and 3.8B students with QLoRA, compile to int4 ONNX, and evaluate offline against a hash-frozen holdout and closed-loop in SUMO on measured London topologies over thirty training-disjoint seeds per cell.

Three results stand out. Distillation *saturates at 0.6B*: the 3.8B student is statistically indistinguishable from the 0.6B one offline and closed-loop, and the two students' catastrophic failures coincide on the same seeds. Closed-loop delay *decouples* from decision accuracy — a 42-point accuracy gain yields no significant delay improvement inside the shield's envelope. A three-seed finding that MaxPressure is worse than a naive fixed-time floor on real London topologies fails to replicate at thirty paired seeds, and I report that retraction rather than the headline it would have made. What fine-tuning buys instead is *authorship*: the model's share of all served actuations rises from 0.39–0.62 to 0.56–0.70, because the distilled policy trips the anti-starvation floor far less often, while its divergence from MaxPressure among the decisions it authors falls from 0.46–0.60 to 0.11–0.18. That matters for accountability: an audit of the stock deployment could attribute two decisions in five to the learned component, against four in seven after distillation, a deterministic timer authoring the rest. I also state the deflationary reading this invites — that distillation partly taught the student to imitate the classical reference — and bound it with the available evidence. Seven collision scenarios reconstruct end-to-end with tampering caught, an adversarial trust battery shows blatant misreporting contained but stealth misreporting unresolved, and the nulls, latency exclusions, and specialisation costs are reported with their mechanisms rather than omitted.

Acknowledgements

I thank Dr Akin Delibasi for supervision that consistently pushed this work towards harder evidence — the throughput-alongside-delay test, the multi-seed robustness requirement, and the scope expansion into fine-tuning all originate with him. I thank Lee Stott (Microsoft) for the policy-driven architecture guidance that shaped the authorisation layer, and specifically for the objection that became §6.9: that a model cannot be judged on a task whose rules were never put in front of it. Testing that objection properly changed what this dissertation is able to claim. I am grateful to UCL Computer Science for access to the departmental GPU cluster, on which the 3.8B student and its five ablation variants were trained.

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Chapter 1

Introduction

1.1 Motivation

Traffic signals are the most consequential actuators most cities own, and the research frontier has begun handing them to large language models: GPT-4-distilled controllers [13], RL-tuned reasoning models [36], and cooperative LLM agents [32] now populate the literature. Yet every one of these controllers is evaluated against a cloud endpoint, while the junction itself is an edge environment: a signal cabinet has no guaranteed backhaul, a hard memory envelope, and — uniquely among LLM application domains — a statutory duty to account for itself when a collision occurs. Between the literature and a deployable junction sit two unanswered questions: whether a language model small enough to live in the cabinet can control real junctions competently, and whether its decisions can be bound into a record an investigator can trust. This dissertation answers both together, because they are not separable: an accountability record is only meaningful if the component it certifies actually authored the behaviour.

The work is conducted with and for a Microsoft-supervised context, which fixes the serving stack as a hard constraint: Microsoft Foundry Local on a 6 GB consumer GPU, capping the model ladder at roughly 4B parameters. I treat the constraint as the research object rather than an inconvenience — the question is what *this* deployable envelope can do, not what a datacentre could.

1.2 Research questions

RQ1 Can an on-device SLM (≤4 GB, Foundry Local, RTX 2060-class), behind a deterministic safety shield, match or beat classical baselines (fixed-time, MaxPressure) on measured London topologies in SUMO?

RQ2 What does distilling a frontier teacher into the SLM buy, offline and closed-loop — and at what model scale does the benefit saturate?

RQ3 Can every actuation be bound into a tamper-evident audit that, after a simulated collision, yields a UK-law-grounded evidence pack rather than an unaccountable log?

RQ4 Does a trust-coefficient mechanism over neighbour reports withstand honest, blatant, and stealth misreporting?

1.3 Contributions

1. **The first documented end-to-end chain** from QLoRA fine-tuning through int4 ONNX compilation to Foundry Local serving of a custom model, with bit-identical CPU/GPU decision equivalence and four documented serving-reliability findings with mitigations (§4.4, §6.10).

2. **A scale-saturation result, at two levels**: frontier distillation lifts a 0.6B student to the teacher-agreement ceiling (97–99%) and a 3.8B student lands on statistically identical numbers, offline and closed-loop (§6.3); moreover the two students fail on the *same* seeds with matched magnitudes, so what they share is not merely an accuracy ceiling but a policy attractor (§6.6). Capability is not the bottleneck at this task — the distillation ceiling is.

3. **The authorship finding**, the dissertation's headline: offline accuracy decouples from closed-loop delay (neither student significantly beats the fixed-time floor at n=30), but fine-tuning raises the SLM's share of all served actuations from 0.39–0.62 to 0.56–0.70 by tripping the anti-starvation floor far less often, while cutting its divergence from MaxPressure from 0.46–0.60 to 0.11–0.18.

Authorship, not delay, is what distillation moves — reported at its true magnitude, with the rival imitation reading stated (§6.5).

4. **A replication failure reported against my own claim**: an exploratory three-seed sweep showed MaxPressure worse than naive fixed-time on every real London topology; at thirty paired seeds the effect does not hold — MaxPressure is significantly *better* on Old Street, tied on Bloomsbury, and worse only in an uncertain mean on the Euston corridor. I retract the claim and replace it with the narrower supported one: MaxPressure's advantage over a naive plan is unreliable on irregular real geometry, though uniform on synthetic grids (§6.1).

5. **A proven audit-to-law pipeline**: seven collision scenarios reconstructed from signed, hash-chained records, with tampering caught and the governing UK rule inferred deterministically from verified facts, emitting cited evidence packs rather than verdicts (§6.7).

6. **An honestly-scoped trust mechanism, tested against the controller that matters**: asymmetric, discount-only trust that contains blatant misreporting but is evaded by a calibrated stealth attacker, and — run against fine-tuned as well as stock controllers — the finding that authorship cuts both ways: a model that genuinely acts on its coordination inputs is moved further by poisoned ones (§6.8).

7. **Two negative results established under fair protocols**: an authorisation re-test in which explicit in-prompt policy rescues a stock model's recall but not its safety, while the fine-tuned students prove format-captured on the adjacent task (§6.9); and the measurement that serving-artifact provenance alone shifts a headline delay result by nine points, which is why every comparison here uses an identical-pipeline control (§6.2).

1.4 Scope and thesis statement

The locked thesis question of this project is exploratory: *is an on-device Foundry-Local SLM traffic controller that matches or beats classical control on real London topology, while producing a provable accident audit, possible at all — and at what model scale and configuration?* It is a feasibility and scale-threshold study. Everything runs in SUMO; three real London topologies plus synthetic grids; demand magnitudes are measured, their realisations synthetic; comparators are classical (fixed-time, MaxPressure), not trained deep-RL systems, because the contribution is on-device feasibility plus auditability, not a new state of the art in delay. Limitations are reported as findings throughout — the evaluative stance is that a negative with a mechanism is worth more than an unexamined positive.

1.5 Dissertation structure

Chapter 2 positions the work against the classical, learning-based, and LLM signal-control literatures and the audit and shielding theory it builds on. Chapter 3 presents the architecture and defines the authorship metrics. Chapter 4 records the engineering, including the serving chain. Chapter 5 fixes metrics, statistics, and leakage controls. Chapter 6 reports the evidence in the order it was built. Chapter 7 answers the research questions, analyses the delay-decoupling mechanism, and states threats to validity. Chapter 8 concludes.

Chapter 2

Background and related work

2.1 Classical signal control and its guarantees

[evidence: ANALYSIS.md \S 3 (backpressure cluster)]

Deployed urban signal control still rests on two families. Fixed-time control serves a pre-computed cycle whose stage splits are engineered offline from historical counts; it optimises nothing online, but it is deterministic, starvation-free by construction, and indifferent to sensing noise. Its adaptive rival, MaxPressure [26], is the field's canonical strong baseline: each intersection independently serves the phase with the greatest "pressure" (upstream minus downstream queue lengths, weighted by turn ratios), and Varaiya proved this decentralised rule throughput-optimal — it stabilises every stabilisable demand. That guarantee, however, is bought with assumptions, and a critical reading of the backpressure literature shows that in central London each assumption fails structurally rather than marginally.

First, the optimality proof assumes unbounded queue storage. Gregoire et al. [6] prove unconditionally — for any topology — that linear-pressure control of exactly Varaiya's form ceases to be work-conserving once queue capacities are finite, and they construct explicit deadlocks; their repair, capacity-normalised pressure, is *not* part of standard MaxPressure. Central London's short inter-junction links mean small capacities everywhere, so the theorem's failure condition is standing, not exceptional. Second, the only stability result for a deployable (non-oracle) backpressure controller holds under heavy load only: Zaidi et al. [33] require every queue to exceed saturation flow, concede that sub-saturation queues "can unstabilize the queuing network", and describe stability outside heavy load as "still a challenging problem". Measured diurnal demand sits below that threshold for most of the day, precisely the regime in which the pressure signal chases arrival noise while a fixed cycle holds steady. Third, every favourable empirical figure in this lineage is a grid figure — roughly 90% of oracle backpressure on a uniform 21×21 grid, falling to about 80% under heterogeneous parameters — accompanied by the authors' own caveat that grid results cannot be extended to arbitrary networks. Fourth, junctions with give-way movements, gap acceptance, or gyratory circulation fall outside the model domain entirely: service in these theorems is binary and phase-determined, so no guarantee applies at all.

The field's own tables corroborate the predicted direction. AttendLight [19] reports MaxPressure *losing* to naive fixed-time by 5–19% on its least regular topologies (T-junction configurations) while winning on regular four-way grids — degradation strongest exactly where geometry departs from the grid. Against this stands a mandatory scope boundary: at least six independent studies [7, 13, 28, 31, 32, 36] show MaxPressure beating fixed-time by 7–58% on CityFlow grid-like networks, including nominally "real" Manhattan, Jinan, and Hangzhou topologies — but always under synthetic or replayed-uniform demand. My own measurements (§6.1) sit between these positions: at thirty paired seeds MaxPressure is significantly better than fixed-time on my most complex real junction, tied on a second, and worse only in an uncertain mean on a saturated corridor, while behaving conventionally on my synthetic grids — which both pre-empts the objection that the baseline is broken and withholds the stronger reversal claim an earlier three-seed sweep had suggested. The gap this dissertation fills is that no prior work empirically tests MaxPressure against fixed-time on measured UK geometry under measured demand, still less connects the outcome to the theorems above.

2.2 Learning-based signal control

Deep reinforcement learning dominates the academic literature that classical control left behind. FRAP [35] introduced phase-competition invariances; CoLight [28] added graph-attention coordination across intersections; AttendLight [19] pursued a single attention-based policy transferable across intersection layouts; Wei et al. [27] survey the wider lineage. The results are genuinely strong — on the benchmarks the community chose. Those benchmarks are almost uniformly CityFlow grids with synthetic or replayed demand, and reporting is typically single-run, so the literature's headline numbers carry no variance estimates. Two further limits motivated the field's turn to language models: trained policies are opaque, offering an operator no account of *why* a phase was served, and the sim-to-real gap is largely unexamined because the simulators and the demand models are shared across papers — an evaluation monoculture. This dissertation does not compete as an RL method; it inherits the RL literature's problem framing (phase selection on a fixed decision interval) while rejecting its evaluation habits: I use measured topologies and demand, disjoint seeds with confidence intervals, and an architecture whose every decision is recorded with its stated rationale.

2.3 LLMs and SLMs for traffic signal control

[evidence: ANALYSIS.md \S 2 competitor map + \S 8 gaps]

The direct competitors apply language models to signal control, and the map is best read along three axes: where the model runs, how the evidence is powered, and what happens after a decision is made. LLMLight/LightGPT [13] is the closest prior work: GPT-4 rationales distilled into LoRA-tuned students down to 0.5B — but served on four A800-80GB GPUs, evaluated single-run on CityFlow with no seeds or confidence intervals anywhere in the paper. Traffic-R1 [36] trains a 3B model with GRPO and deserves real credit for a ten-intersection production deployment and for an ablation showing SFT degrading the base model's general benchmarks — yet its own future-work section concedes that edge deployment remains unrealised; this dissertation builds the artefact that paper only gestures at. DGLight [31] (a dissertation preprint, not peer-reviewed) tunes an 8B model on dual H100s and usefully calibrates the field: the *entire* LLM-versus-MaxPressure gap it and its peers report is roughly 1–2%, a margin that would not survive the seed variance I measure and that no competitor tests for. CoLLMLight [32] scales to 196 intersections on server GPUs and, tellingly, finds its fine-tuned 8B model beating a stock 671B model — distillation beating scale, which supports my central bet. CuraLight [7] is the one fine-tuning work on SUMO, but its 12B model needs roughly 44GB of VRAM against my 6GB budget. iLLM-TSC [20] inverts my safety topology — an RL agent proposes and a cloud GPT-4 vetoes — on a single synthetic four-way intersection, with a retry loop that falls back *silently* to the unvetted RL action; I return to this in §2.5, and I acknowledge its degraded-communications threat model as a scenario axis I do not test. LATS [34] distils an LLM teacher into MLP/GRU students — so nothing generative is ever deployed — and its pure-LLM baselines perform worst of all its methods, corroborating my finding that stock small models are weak deciders.

The pattern across all seven is consistent: every system serves from cloud or server GPUs; every headline table is single-run or, at best, $n \leq 10$ without statistical tests; none pairs control with any audit mechanism; and none evaluates on measured UK topologies. Those four absences are, jointly, the opening this dissertation occupies.

2.4 Small language models, distillation, and quantization

[evidence: ANALYSIS.md \S 4–\S 5]

I adopt the framing of the SLM survey [18] and the on-device review [29]: this system is *edge-only* (no cloud fallback), and I report the review's indicator template (time-to-first-token, VRAM, latency percentiles). Strikingly, neither survey contains a single traffic or control-loop case study — the application gap exists by omission. The nearest on-device comparator is Octopus [2], a 2B function-calling model at 1.1–1.7s on Android, bracketing my 0.3–1.5s decision latencies. The surveys also corroborate a reliability floor: Qwen2.5's own report [22] shows instruction-following (IFEval) collapsing from 84.1 at 72B to 27.9 at 0.5B, consistent with the 45–69% decision accuracy of my stock models. The Qwen3 report [23] complicates this — its 0.6B model posts respectable IFEval scores yet was my weakest stock decider; I resolved this in-house (the model parsed 93–100% strict JSON while choosing wrong phases), which sharpens rather than undermines the point: generic instruction-following benchmarks do not measure numeric decision quality, and the technical reports never test the latter.

On distillation, I must disclose a tension. Orca [17] argues that small models need rich explanation traces, whereas my distillation is label-only. Three defences: the task is a bounded, closed decision schema — closer to classification than open generation — and the knowledge-distillation survey [30] endorses labelling-plus-SFT for exactly this cell; Orca itself concedes small models excel in constrained settings; and empirically my student reaches near the teacher ceiling (97.7% against 98–99% teacher self-agreement). I concede what Orca implies: out-of-distribution generalisation and natively generated audit rationales are untested risks. The jump from 55% to 97.7% has precedent — distilled students outperforming their data-scale expectations [9, 25]. For the training recipe, LoRA [10] and QLoRA [4] justify rank 16, though their evidence stops above 125M parameters, and I flag the extrapolation. One precision matters: QLoRA trains adapters on a 4-bit base and serves unmerged; I train QLoRA-style, merge to bf16, then apply post-hoc int4 round-to-nearest for serving — two different quantisation steps, and QLoRA's authority does not transfer to the second. The quantisation literature in fact predicts trouble there: GPTQ [5] shows small models are the hard case for round-to-nearest, AWQ [16] finds the gap narrowing at 4-bit with grouping, and post-hoc quantisation after fine-tuning has been measured losing 9.4% relative even with GPTQ [24]. My contrary finding must therefore be scoped narrowly: bit-identical held-out decision accuracy under weight-only int4, on a near-converged narrow-schema checkpoint, measured with a discrete accuracy metric coarser than perplexity — a favourable open finding, not a literature-endorsed property. Finally, forgetting-scaling results [12] predict general-capability drift at my very low training loss, and show task accuracy and base-behaviour drift can diverge; I carry this as a declared limitation, doubly motivated by Traffic-R1's forgetting ablation. The gap this dissertation fills: no published work provides per-decision reliability data for a generative model at 0.6B, int4-quantised, inside a closed control loop — the surveys name size-versus-reliability as unresolved, and my measurements are new evidence on it.

2.5 Safe-by-construction shields

The canonical formalism is shielding [1]: a reactive system synthesised from an LTL safety specification and a finite MDP abstraction by solving a two-player safety game, placed either preemptively (the shield emits the safe-action set) or post-posed (the shield substitutes unsafe chosen actions). It carries proofs my system does not: correctness for all abstraction-consistent environments, minimal interference, and preserved learner convergence — and the authors' warning that the shield must remain active after learning, which mine does. Below this sit two rungs. Invalid-action masking [11] proves the masked update is a valid policy gradient and shows masking scales where penalties fail; agents trained masked remain largely valid unmasked. Neurosymbolic masking [8] learns the symbolic grounding when constraints are unknown — and its own future work points back towards temporal-logic specifications.

Positioned honestly, my shield is *not* a shield in the Alshiekh sense. It composes three weaker but sound pieces: preemptive action masking (the SLM selects from a fixed menu of pre-vetted conflict-free phase programs, so "no conflicting greens" holds by construction of the action set); a post-posed default-substitution filter (silent or invalid proposals are replaced by the deterministic MaxPressure choice); and an anti-starvation floor enforced by code rather than proved from a specification. What is missing is the LTL specification, the abstraction, and the game — hence no minimal-interference proof and no lookahead: my floor reacts *at* the starvation threshold where a synthesised shield acts before violation becomes unavoidable. The upgrade path is concrete and cheap at single-junction scale, and I name it as future work. This placement also explains why my topology beats iLLM-TSC's inversion [20]: they seat the least verifiable component (a prompted cloud LLM) in the safety-monitor position and fall back silently when it fails to parse, whereas I put the unverifiable component on the proposal side and keep the veto deterministic, local, and auditable — the only arrangement compatible with a 6GB edge premise. The gap: no prior LLM-based signal-control work states the formal status of its safety layer at all; this dissertation does, including what its shield is not.

2.6 Audit, transparency, and legal grounding

[evidence: 01-research/uk-traffic-law.md; ANALYSIS.md \S 7]

The dissertation's headline thread — that an edge controller should produce a provable account of its own decisions — sits in a three-way gap in the trust literature. On the attack side, Qu et al. [21] show that roughly 6% of colluding vehicles feeding falsified data to an undefended learning controller cut the colluders' waiting time by 92.5% while raising everyone else's by 62.7%; their most dangerous result is that a *calibrated* lie beats a maximal one, which directly motivates my stealth-liar scenario, and their paper explicitly names real-time anomaly detection as unbuilt future work. On the integrity side, the blockchain architecture of [15] rejects 100% of spoofed inputs at ~39ms — but never names a signature scheme, and its own future work concedes failure when the validators themselves collude; more fundamentally, input integrity is not decision accountability: it certifies what the controller *heard*, not what it *decided* or why. Nobody in this literature builds the defence.

My audit design therefore borrows its mechanism from outside traffic: Certificate Transparency [14] demonstrates that an append-only Merkle log with inclusion and consistency proofs makes tampering detectable without trusting the log operator — exactly the property a post-accident investigation needs, at a per-decision cost (a signature and a hash) that fits the edge budget where a validator network does not. The legal layer is deliberately modest. Dahl et al. [3] profile pervasive hallucination when language models answer legal questions, which rules out letting any model — let alone a 0.6B one — assert fault. My system instead emits a *cited evidence pack* against a UK-law fault-attribution knowledge base (§3.4): under it, authority-side liability is legally weak and arises essentially only from conflicting-green misfeasance, driver-fault doctrines are strong, and emergency-vehicle exemptions are conditional — so the artefact's job is to preserve the signed decision record that lets human lawyers apply those doctrines, never to output a verdict. The gap this dissertation fills is the strongest of the five it claims: it is the only system in this literature with a decision-level cryptographic audit trail, the only defence-side trust mechanism (a ledger in which truth-telling raises trust and detected lies are heavily penalised, tested against honest, blatant, and stealth adversaries), and consequently the only one that can produce a crash-to-UK-law evidence pack at all.

Chapter 3

System design

The system's design premise is that a learned controller at a junction must be *optional for safety and mandatory for accountability*: the deterministic control path must be complete without the model, while every actuation — whoever authored it — must be provable afterwards. This chapter presents the architecture that realises the premise, and defines the three structural quantities (authored share, proposal acceptance, divergence rate) whose measurement becomes the dissertation's central finding.

3.1 Architecture overview

Each junction runs an identical control loop on a fixed decision interval ($\approx$10 s simulated). Per tick: the junction's local state (per-phase queue and waiting measurements from its own sensors) is rendered into a prompt; the on-device SLM, served by Foundry Local, proposes a phase; a deterministic parser extracts the proposal; the safety shield masks, validates, and possibly substitutes it; the served phase actuates the lights; and a signed audit record binding state, proposal, served action, and authorship is appended to the junction's hash chain. The MaxPressure policy is computed every tick regardless of the SLM — it is simultaneously the shield's reference, its fallback, and the strong baseline — so the marginal cost of the learned component is one model call, and its marginal risk is bounded by construction.

Three design consequences follow. First, graceful degradation is the default: if the model times out, emits garbage, or the serving runtime dies (all observed in practice, §6.10), the junction continues under MaxPressure with the failure recorded. Second, the architecture inverts iLLM-TSC's topology [20] — there, an RL agent proposes and a cloud LLM vetoes; here, the LLM proposes and a local $O(\text{phases})$ verifier vetoes — which is the only orientation deployable at the edge, since the veto path must be cheap, local, and always available. Third, because the deterministic path is complete, every claim about the SLM's value can be measured as a *difference* against the same loop with the proposer disabled.

3.2 The SLM decision agent

The agent contract is deliberately minimal: the model receives a system instruction and a rendered state, and must return exactly `{"phase": N}`. Temperature is zero; the serving endpoint is OpenAI-compatible; parsing falls through four tiers (strict JSON, braced, keyed, bare integer), with the tier recorded per decision so format discipline is measurable, not assumed.

Four prompt formats act as graded capability probes, each revealing a different competence. **Myopic** exposes only per-phase halting counts — the least privileged view, answerable by a counting heuristic. **Sota** adds delay-aware context (accumulated waiting), the state a delay-optimal policy actually needs. **Prediction** adds short-horizon arrival forecasts, probing whether the model can use anticipatory information. **Coordination** adds neighbour notes, probing multi-junction awareness. The formats are held identical between offline evaluation and closed-loop control — the same rendering code serves both — so offline accuracy numbers transfer interpretably to the loop.

3.3 The safety shield

The shield composes three mechanisms, named here with the vocabulary of the formal literature (§2.5) so that what it is *not* is explicit:

1. **Preemptive action masking** [11]: the model chooses from a fixed menu of conflict-free phases; no proposal can create a conflicting green, because conflicting actions are not in the action space.
2. **Post-posed default substitution**: an invalid, unparseable, or absent proposal is replaced by the

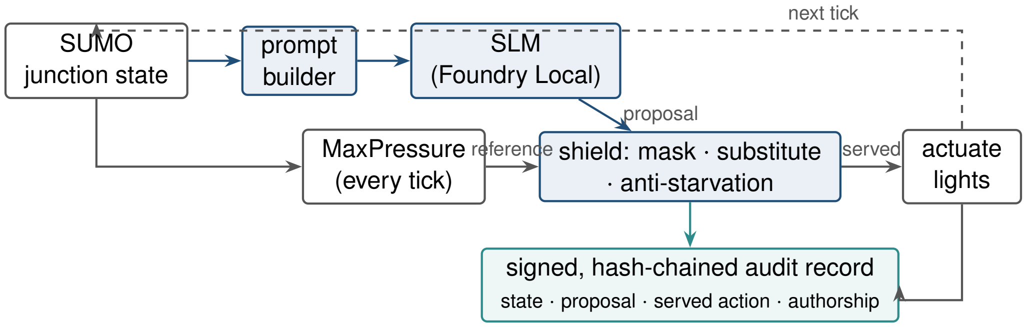

Figure 3.1: The junction control loop. The deterministic path (grey: state → MaxPressure → shield → actuation) is complete without the SLM; the learned proposer (blue) attaches to it, and every served decision — whoever authored it — lands in the signed audit chain (teal).

MaxPressure phase. The substitution is recorded as such — it is a served decision the SLM did not author.

3. **Anti-starvation runtime monitor**: a minimum-service floor guarantees every approach is served within a bounded window, overriding both the SLM and MaxPressure if necessary.

This is a sound heuristic composite, not a correct-by-construction shield in the Alshiekh sense [1]: there is no temporal-logic safety specification, no game-theoretic synthesis, and no minimal-interference proof. I state this as a limitation with an upgrade path (§7.4); the composite nonetheless guarantees the two properties the evaluation depends on — no conflicting greens, no starved approach — for every arm including the baselines.

Three measured quantities are defined on this structure, and their denominators matter because Chapter 6 turns on them. Every decision tick is labelled by *who authored the phase actually served* — the SLM, the MaxPressure substitution, or the anti-starvation floor — and the three metrics are taken over different subsets of those ticks.

Authored share is slm/decisions: the fraction of *all served actuations* whose phase originated with the SLM. This is the quantity the audit argument needs, because it is the share of deployed behaviour attributable to the learned component.

Proposal acceptance is slm_served/slm_valid_proposals: of the ticks where the SLM returned a well-formed proposal, the fraction where that proposal was the phase served rather than being displaced — in practice, displaced by the anti-starvation floor, since substitution is rare.

Divergence rate is diverged_and_served/slm_served: of the decisions the SLM authored, the fraction whose phase differed from what MaxPressure would have chosen. It measures how far the served policy departs from the classical reference — *not* how often the shield vetoed the model, which masking and the rarity of substitution together make close to never.

None of the three is a count of unsafe actions blocked: masking makes unsafe proposals unrepresentable, so the shield's substitution path only catches malformed or absent output. Reporting all three separately is what prevents the interpretive error of reading a high acceptance rate as a high share of authorship (§6.5).

3.4 Audit trail and crash reconstruction

The audit layer is designed backwards from the question an investigator asks after a collision: *what did the controller know, what did it decide, who authored the decision, and can the record be trusted?*

Every decision tick appends a record containing the state digest, prompt hash, model identity and version, raw proposal, parse tier, served action, authorship (SLM / shield-substitution / anti-starvation), and latency. Records are Ed25519-signed and hash-chained per junction; chain verification and signature verification run end-to-end after every experiment, and the threat model includes an imposter re-signing attack (caught in all evaluated scenarios, §6.7). The chain head can be anchored to external witnesses in a Certificate-Transparency-style quorum [14]; a permissioned-ledger anchor is implemented as one optional witness, deliberately positioned as an anchoring option rather than a contribution.

On top of verified records sits the reconstruction layer: a UK-traffic-law knowledge base (statutes, case law, the conditional emergency-vehicle exemption) against which the verified facts of a collision scenario are matched deterministically. The output is a *cited evidence pack* — the governing rules, the verified facts that trigger them, and the fault anchor each rule implies — never a verdict. This design choice is a direct response to the legal-hallucination literature [3]: the system's legal knowledge is a curated, citable database, and the adjudicative act is explicitly left to humans. The dissertation's own scenarios use synthetic actors on real topology, so no real data subject exists; the residual legal exposure of a real deployment is analysed in §7.3.

The audit design is also why authorship (§3.3) is the pivotal metric: an evidence pack over a deployment whose decisions are mostly shield substitutions attributes the behaviour to a textbook algorithm, not to the certified learned component. For the audit to be *about the model*, the model must author the decisions — which is precisely what fine-tuning turns out to buy (§6.5).

3.5 Trust-coefficient mechanism

Coordination requires acting on neighbour reports, which is exactly what a compromised neighbour can abuse. The trust mechanism prices that risk with three commitments. First, **local sensing is the ultimate truth**: a junction's own sensors are never doubted, and every neighbour claim is eventually scored against what local sensing later verified. Second, **asymmetric update**: verified truths earn trust slowly (prior 0.5, ten consecutive truths to reach 0.9) while a detected lie from a high-trust source collapses it in one step (lie factor 0.25), because in infrastructure the cost of trusting a liar exceeds the cost of doubting an honest peer. Third, **discount-only wiring**: trust can strip a caught liar's power to corroborate a preemption, but can never *add* an action on zero local evidence — so the mechanism cannot regress safety, and a real ambulance is always preempted via the junction's own sensing. A truthful report that local sensing verifies late is vindicated retroactively, so honesty is never permanently penalised for latency. The adversarial evaluation (honest, blatant, stealth misreporting injected into closed-loop SUMO) and its honest partial results are reported in §6.8.

3.6 The fine-tuning axis

The final design axis asks what changes when the proposer is fine-tuned rather than stock. The pipeline is distillation: a frontier teacher labels junction states sampled from training-reserved seeds; labels are cross-audited against a second frontier model (98–99% agreement); a deterministic sanity verifier screens states, not answers, so the teacher is never second-guessed by a weaker rule; and the four prompt formats are rendered from the same labelled pool into one mixed training set. Training is QLoRA [4] (r=16, NF4 double quantisation, completion-only loss, family-correct chat templates); the merged bf16 model is compiled to weight-only int4 ONNX and served through the same Foundry Local runtime as every stock model.

Three design rules protect the comparisons. The holdout is hash-frozen before any training (§5.3); closed-loop evaluation seeds are disjoint from training seeds; and the control arm is the same base model compiled through the identical int4 pipeline, so the only manipulated variable is the weights. Ablation students — leave-one-topology-out and four format specialists — are trained under the same recipe to test memorisation and specialisation hypotheses directly rather than by argument.

Chapter 4

Implementation

This chapter records the engineering that made the design measurable: the simulated London networks, the on-device serving stack and its documented failure modes, the distillation and compile chain, and the experiment harness. I keep it tight and evidence-focused; the reproducibility pins (versions, seeds, templates) are tabulated in Appendix A.

4.1 Simulation environment and topologies

All control runs in SUMO with TraCI. Six topologies span the regular-synthetic to irregular-real axis that Chapter 6 shows is decisive: two synthetic grids (3×3, 4×4); the Euston A501 corridor and its peak-hour variant, whose demand uses the measured DfT hourly diurnal profile and vehicle mix for the A501; a Bloomsbury grid extracted from OpenStreetMap; and the Old Street complex junction (nine signals including a single five-arm, seventeen-movement junction, demand-calibrated after default random demand gridlocked it). Real-map demand magnitudes are measured; their vehicle-by-vehicle realisations are synthetic draws per seed — a scoping honestly carried through every claim (§7.3).

4.2 On-device serving with Foundry Local

Serving is Microsoft Foundry Local (pinned 0.8.119) on an RTX 2060 with 6 GB VRAM — consumer hardware of roughly signal-cabinet class, and a hard project constraint rather than a convenience: the model ladder is capped at ~4B (7B-class models OOM the runtime, recorded), which is what makes the scale-saturation result (§6.3) deployment-relevant rather than academic.

Custom models serve through two execution providers. The CPU path works out of the box; the WebGPU path requires editing the compiled model's `genai_config.json` provider options — and delivers roughly 2× lower latency with *bit-identical* holdout decisions, which licensed running every sweep GPU-served. That speedup is measured against the same artifact on CPU and is not a claim of absolute speed: §6.10 reports my pipeline-compiled 0.6B artifacts serving at roughly twice the median latency of the vendor catalogue build of the same weights — a real cost of controlling the compile chain, and the reason the 3.8B student is the fastest arm I serve. Windows ML CUDA autoregistration requires a newer Windows build than the test machine's, making WebGPU the effective GPU path; I document this because none of it is in the product documentation, and the recipe is part of the contribution.

Sustained unattended serving surfaced four reliability findings — catalogue-artifact token corruption after long uptime, two unprompted service self-restarts mid-evaluation, and undocumented registration requirements for compiled models — each mitigated in harness (§4.5) and reported as results (§6.10).

4.3 Distillation dataset and training

From four training-reserved seeds I sampled 5,952 unique junction states and labelled them with a frontier teacher; a second frontier model re-labelled audit samples at 98.0–99.0% agreement. Rendering the four prompt formats over the labelled pool yields 59,533 training examples and a hash-frozen holdout of 5,580 (`HOLDOUT_MOD=12`). Rendering is family-aware — ChatML with an empty think block and no-think marker for Qwen3; Qwen2.5's ChatML without the marker; Phi's `<|system|>...<|end|>` convention — with loss computed on the completion only, targeting exactly `{"phase": N}`.

Training is QLoRA r=16, NF4 double quantisation, fp16 compute. The 0.6B student trained on the serving machine itself (13.5 h on the RTX 2060 — the full loop from teacher to served controller fits on one consumer device); the 3.8B student and its five ablation variants trained on a single UCL RTX 4090

(2.3 h each, 8.9M trainable parameters, 0.23%).

4.4 Compile-and-serve chain

The path from merged bf16 weights to a served endpoint is undocumented for custom models, and assembling it produced pinned lessons now recorded in the project's feasibility log: the onnxruntime-genai model builder must run against a specific transformers pairing (newer tokenizer configs emit a `tokenizer_class` Foundry cannot load — patched to the catalogue value); Phi checkpoints need their remote-code mappings stripped; and a compiled model is not servable until an inference manifest (`inference_model.json` — name and prompt template) is *synthesised*, because the builder does not emit one. The eventual chain — QLoRA merge → int4 RTN compile → config patch → manifest synthesis → cache registration → WebGPU provider swap — is, to my knowledge, the first documented end-to-end recipe for serving a custom fine-tuned model through Foundry Local.

4.5 Experiment harness

Two harnesses produced every number in this dissertation. The **factorial runner** executes model × format × topology × seed cells with per-cell raw JSON artifacts, a pre-flight serving probe per cell (no silent parity: an unservable model records a skip, never a score), resumability (existing cells are skipped, so crashes and restarts lose at most one cell), and multi-pass retry for deferred cells. The **offline evaluator** scores holdout rows against the served endpoint with fail-loud error accounting — API errors are counted, never scored as answers — and, after the observed service self-restarts, recovery logic that rediscovers the endpoint, reloads the model, and retries failed rows with statistics intact. The harness survived a six-week unattended campaign including two service restarts, one machine reboot, and one mid-write power loss (one corrupted cell, detected by JSON validation and re-run); I regard the fail-loud discipline itself as a methodological contribution of the work.

Chapter 5

Experimental methodology

This chapter defines what I measured, how each comparison was controlled, and the statistical decision rules I committed to before running the experiments. The methodological stance throughout is fail-loud: harness errors abort or are counted, never silently scored, and null results are reported as findings.

5.1 Evaluation axes and metrics

I evaluate along four axes, each with its own primary metric.

Closed-loop control quality. The primary metric is mean network delay: the mean, over all vehicles loaded into the simulation, of the excess travel time relative to free-flow, measured over a fixed 1,200-step horizon. Every controller cell is paired with a fixed-time baseline run on the *same topology and seed*, so the reported quantity is the paired relative delay change. Secondary metrics — completed-vehicle throughput, teleport count, and undeparted backlog — guard against the classic failure mode of delay metrics, where a controller "improves" delay by starving demand.

Offline decision quality. Accuracy against frozen frontier-teacher labels on a held-out set of junction states, at int4 precision exactly as served. I also report strict-JSON rate (the fraction of responses matching `{"phase": N}` exactly) and parse rate (any recoverable phase), because a controller that cannot keep its output contract imposes a different kind of cost than one that chooses a wrong phase.

Structural authorship. The three quantities defined in §3.3 — authored share (SLM-served ticks over all decisions), proposal acceptance (SLM-served over SLM valid proposals), and divergence rate (served decisions differing from MaxPressure, over SLM-served) — reported alongside the decision composition itself (SLM / substitution / anti-starvation). I treat these as first-class results rather than diagnostics, because the audit argument of this dissertation (§3.4) turns on the first of them, and because reporting acceptance without authorship invites exactly the misreading that a high acceptance rate implies the model runs the junction.

Serving viability. Decision latency (p50/p99) against the 10 s decision interval, and a log of serving-reliability events with their mitigations.

5.2 Statistical protocol

Every closed-loop cell is replicated over n=30 random seeds per topology. The test for a delay effect is a paired t-test on per-seed (controller, baseline) delay pairs; I additionally pool across topologies (paired, n=90) where the hypothesis is topology-independent. p-values are computed with the project's pure-mathematics implementation of the regularised incomplete beta function — the same code path used by the two-way ANOVA of the explanatory study — so no statistics library version can silently change a reported number. For the factorial explanatory analysis I report effect sizes (η^2) alongside F statistics, and for headline comparisons I state the direction, magnitude, and p together rather than significance alone.

Two commitments were made in advance. First, the honest-null policy: a non-significant result at n=30 is reported with its magnitude and uncertainty, not suppressed or re-run until significant. Second, no mid-experiment stopping: cell counts were fixed by the factorial design before launch, and the resumable runner (§4.5) re-runs only cells that produced no measurement (crash, probe failure), never cells whose answer was unwelcome.

5.3 Seed-disjointness and holdout freezing

Two leakage paths exist between training and evaluation, and each is closed by construction.

State-level leakage. The distillation dataset is built from junction states sampled on four reserved seeds ($\{3, 7, 11, 29\}$). The offline holdout is split from the same state pool by a deterministic hash of the state content (`HOLDOUT_MOD=12`, giving 5,580 of 65,113 rendered examples), and was frozen — file hash recorded — before the first training run. No training example, under any format rendering, shares a state with a holdout example.

Trajectory-level leakage. Closed-loop evaluation uses seeds $\{1..30\} \setminus \{3, 7, 11, 29\} \cup \{31, 32, 33, 34\}$ — thirty seeds fully disjoint from the training seeds. This matters because a controller fine-tuned on states from seed s has effectively seen a compressed replay of seed s's demand realisation; evaluating on it would overstate generalisation. All closed-loop numbers in Chapter 6 are disjoint-seed numbers.

5.4 Baselines and controls

Fixed-time is the floor every deployed junction already achieves: a cyclic plan with equal splits, the same phase menu as every other arm. **MaxPressure** is the theoretically-grounded adaptive baseline [26], and doubles as the shield's fallback policy.

The control for every fine-tuning comparison is the *same base model compiled through the identical int4 pipeline* (denoted stock, or ft0), not the vendor catalogue artifact. This choice was forced by measurement: the catalogue Qwen3-0.6B GPU artifact was observed emitting corrupted tokens after long service uptime (a documented reliability finding, §6.10), and a catalogue artifact differs from my compiled one in quantisation recipe even when healthy. Holding the compilation pipeline fixed isolates the single manipulated variable — the fine-tuned weights.

One model class is excluded by measurement rather than swept: phi-4-mini-reasoning's p99 decision latency (9.7 s) consumes the entire 10 s decision interval on this hardware, so closed-loop cells would measure a controller acting on stale state; it is recorded as latency-excluded (its single corridor probe is retained descriptively) — itself a deployability finding about reasoning-class models at the edge.

Two further controls target specific hypotheses. A **leave-one-topology-out** (LOTO) student, trained with all Old Street states removed, tests whether distillation gains are topology memorisation. Four **format-specialist** students, each trained on one prompt format alone, test whether a deployed junction would prefer a per-format model over a generalist; the answer (§6.3) was no at both model scales.

5.5 Experiment inventory

Table 5.1 maps every experiment family reported in Chapter 6 to its design and artifact trail. All raw cells, per-seed JSON records, and aggregation scripts are retained in the project repository; every number in this dissertation is recomputable from them.

The full campaign spanned three machines (the RTX 2060 serving/measurement node, a UCL RTX 4090 for 3.8B training, and an RTX 4060 reserve) and roughly six weeks of wall-clock compute; the factorial policy — capture every cell, gate nothing on interim results — follows the project's standing directive that the dissertation should be able to answer questions I had not yet thought to ask when the sweeps were launched.

Table 5.1: Experiment inventory. Every family retains per-cell raw JSON.

Family	Design	Primary output
Base-model factorial	8 catalogue SLMs $\times$ 4 formats $\times$ 6 topologies $\times$ up to 32 seeds	closed-loop delay vs baselines
Explanatory (why) study	2×2 factorial ANOVA over model/config on 4 topologies	F, η^2, interaction
Offline distillation	7 phi rows + 4 qwen rows $\times$ 4 formats, frozen holdout	accuracy, JSON rate
Fine-tune closed-loop	ft vs stock $\times$ 3 topologies $\times$ 30 disjoint seeds (both scales)	paired delay, authorship
Decision battery	cloud frontier vs local SLM on identical states	agreement, latency
Accident / crash-law suite	scripted collision scenarios $\rightarrow$ audit reconstruction	evidence packs, chain verification
Trust battery	honest / blatant / stealth misreporting	detection, delay impact
Emergency / EV suite	EV arrival scenarios per topology	clearance metrics

Chapter 6

Results

I present results in the order the evidence was built: the baseline landscape that reframed the project (§6.1), the base-model factorial (§6.2), the offline distillation tables (§6.3), the closed-loop outcome (§6.4), the authorship finding this dissertation headlines (§6.5), the audit and trust results (§6.7, §6.8), and the serving-reliability evidence (§6.10).

6.1 Baseline landscape: what the fixed-time floor revealed

Adding a naive fixed-time floor to every comparison was the single most consequential rigour decision of the project, and it produced a result I have had to correct twice.

In the exploratory sweep (three seeds per cell, daily-resolution demand) MaxPressure — throughput-optimal in theory [26] and dominant in the simulation literature — appeared *worse than naive fixed-time on every real London topology tested* (Euston +15%, Old Street +5%, Bloomsbury +3%) while dominating the synthetic grids (−27% on grid4×4, −19% on grid3×3). That contrast was striking enough to become a headline claim of an earlier draft.

It does not survive replication. Table 6.1 recomputes the comparison paired per seed over the thirty disjoint evaluation seeds, on the same measured-demand topologies used everywhere else in this dissertation.

Table 6.1: MaxPressure vs fixed-time, paired per seed (n=30). Positive = MaxPressure worse. CI is 95% on the mean relative change.

Topology	MaxPressure	fixed-time	rel.	95% CI	(p)
Euston peak-hour	259.6 s	235.6 s	+19.8%	[+3.3, +36.2]	(.21)
Bloomsbury grid	527.9 s	527.3 s	+0.2%	[−1.5, +2.0]	(.90)
Old Street	148.5 s	159.4 s	−6.1%	[−10.8, −1.3]	(**.016**)

At n=30 the picture is not a reversal but a **topology-dependent draw**. On Old Street — the most complex geometry in the study, a five-arm junction with seventeen movements — MaxPressure is *significantly better* than fixed-time, contradicting the exploratory result outright. On Bloomsbury the two are indistinguishable. Only on the saturated Euston corridor does MaxPressure remain worse in the mean, and there the paired test does not reach significance (p=.21) because the per-seed spread is large; note also that the confidence interval on the mean relative change excludes zero while the paired test on absolute delay does not — a divergence caused by skew, and a reminder that on this topology a mean is a fragile summary.

I therefore retract the general claim. What the evidence supports is narrower and still worth stating: MaxPressure's advantage over a naive fixed-time plan is *not* reliable on irregular real geometry under measured demand — it is real on one topology, absent on another, and negative-but-uncertain on a third — whereas on the regular synthetic grids the literature favours it is uniformly and substantially better. The backpressure theory offers a mechanism for why the arterial is its weakest case: the throughput-optimality proof assumes unbounded queue storage, and finite-capacity treatments prove work-conservation losses [6], while the only stability results for a deployable variant hold under heavy load [33] — conditions a saturated short-link corridor violates in both directions. I offer that as a hypothesis consistent with the Euston direction, not as a demonstrated explanation, since the effect it would explain is itself not significant.

The practical consequence for this dissertation is unchanged and is the reason the floor was added: any SLM-vs-MaxPressure comparison on real topologies is confounded by MaxPressure's own variable strength, so every controller claim here is stated against the fixed-time floor, with MaxPressure reported alongside.

6.2 Base-model factorial

The full factorial (seven servable catalogue models — the 6 GB card caps the ladder at ~4B, with 7B-class OOM failures recorded — × four prompt formats × five topologies) established three facts before any fine-tuning.

First, in the exploratory sweep (three seeds per cell, daily-resolution demand) shielded stock SLM control beat the fixed-time floor on every topology tested — Euston corridor −16%, Bloomsbury −2%, Old Street −14% (seed-noisy), grid3×3 −21%, grid4×4 −28%. I flag immediately that this result does *not* survive replication on the measured-demand corridor, and the powered numbers below supersede it; the exploratory sweep is reported because it is what motivated the powered design, not as evidence for the claim. Second, results must be conditioned on topology: pooling across maps is a Simpson's-paradox trap, since corridor wins and early grid blow-ups average into a meaningless mean. Third, per-decision reasoning quality does not predict closed-loop value: on 80 identical junction states, cloud frontier models agree with the delay-optimal heuristic at 92–96% against 51–72% for local SLMs, yet a near-chance local model still won its closed-loop cells. The controller's value is systemic — the SLM, shield, and event gating together — not raw single-junction reasoning. This decision battery is open-loop by construction (the frontier teacher cannot be called from inside the simulation loop) and is stated as such.

The explanatory two-way ANOVA over the factorial confirms the structure: model and configuration interact strongly (interaction F=32, p<0.001, η^2=0.86), with the effect driven by outlier cells rather than a smooth gradient — a further argument against pooled claims.

The powered stock-model replication (n=30 per cell, four catalogue models × two formats, both baselines) sharpens the picture per topology. On **Old Street**, stock SLM control achieves the campaign's strongest powered wins against the fixed-time floor — Qwen2.5-0.5B sota −11.8% (p<0.001), Qwen3-0.6B sota −9.6% (p<0.001), Phi-4-mini myopic −8.6% (p<0.001) — with one arm also significantly beating MaxPressure (Qwen2.5-0.5B sota, −4.9%, p=.014).

On **Euston peak-hour** under the measured DfT demand profile the result reverses, and I report it as measured: every stock arm is **7–20% worse than the fixed-time floor** (+7.4% to +19.7%), significantly so on four of the eight arms (two-sided p between .017 and .053), while remaining statistically indistinguishable from MaxPressure. This is not a null; it is a measured harm, and it refutes the exploratory sweep's corridor win above. Two differences explain the reversal — the powered run uses the measured hourly demand profile rather than a daily-average magnitude, and the peak-hour topology loads the corridor far harder — and the direction of the correction is the uncomfortable one: under realistic peak demand, a shielded stock SLM makes the A501 corridor worse, not better. The fine-tuned arms (§6.4) do not share this deficit, which is the strongest practical argument in this dissertation for distillation.

The replication also yielded a methodological finding I had not planned to measure. On the 26 seeds shared between runs, the *catalogue* Qwen3-0.6B artifact and my identically-prompted, identically-shielded pipeline compile of the same weights differ by nine points of mean relative delay (+4.1% vs −5.2% against the floor) on this high-variance topology — quantisation and packaging provenance alone shift the headline sign. This is precisely why every fine-tuning comparison in this dissertation uses the identical-pipeline compile as its control (§5.4), and it is a caution for the field: "the same model" is not the same controller across serving artifacts.

6.3 Offline distillation: saturation at 0.6B

Table 6.2 gives the complete offline campaign: both student scales, stock versus fine-tuned through the identical int4 pipeline, plus the LOTO and format-specialist controls, evaluated on the frozen 5,580-state holdout as served (WebGPU, int4).

Five findings follow.

Table 6.2: Holdout accuracy (%) against frozen frontier-teacher labels, int4-as-served. All fine-tuned rows and the stock Phi row emit 100% strict JSON; stock Qwen3-0.6B reaches 93.4% on sota.

	myopic	sota	prediction	coordination
Qwen3-0.6B stock	45.6	55.1	63.3	69.3
Qwen3-0.6B generalist ft	**87.9**	**97.7**	**97.6**	**99.0**
Phi-4-mini stock	87.5	59.0	86.0	87.0
Phi-4-mini generalist ft	**88.7**	**97.1**	**97.4**	**99.3**
Phi-4-mini LOTO	88.3	96.8	97.4	99.1
Phi-4-mini sota-specialist	87.9	97.8	89.5	93.3
Phi-4-mini myopic-specialist	87.0	87.9	88.8	91.8
Phi-4-mini prediction-specialist	87.9	91.3	97.3	98.4
Phi-4-mini coordination-specialist	87.4	89.1	94.7	99.3

1. **Scale saturation.** The 0.6B and 3.8B generalists are statistically indistinguishable on every format: the largest gap is 0.8 points (myopic), against a standard error of about 1.2 points on n=1,510, so no two-proportion test approaches significance. Both sit at the teacher-agreement ceiling on three of four formats (97–99% against 98–99% teacher self-consistency); the myopic format is the exception at 88%, for the reason given below. Distillation saturates this decision task at 0.6B; six times the parameters buy nothing per-decision.

2. **Where the gain lands depends on the base.** The 0.6B is lifted between +29.7 (coordination) and +42.3 (sota) points across the four formats. Phi-4-mini, already 86–88% on three formats stock, gains most on the delay-aware sota format (+38.1 points), materially on prediction (+11.4) and coordination (+12.3), and negligibly on myopic (+1.2). For a capable base, fine-tuning buys decision quality, not format discipline — stock Phi is already 100% format-compliant.

3. **Generalists only.** No format specialist beats the generalist on its own format by more than 0.7 points, while off-format costs run from 0.8 to 9.2 points; and the myopic specialist loses *even on-format* (87.0 vs 88.7), consistent with mixed-format training regularising against the noisy privileged-teacher labels that pure-myopic training overfits. This comparison was run at 3.8B only — no Qwen format specialists were trained — so it is a single-scale result, unlike the generalist comparison above.

4. **No topology memorisation.** The LOTO student, trained with all Old Street states removed, matches the generalist on every format.

5. **Quantised serving is exact for this task.** CPU and WebGPU int4 artifacts agree bit-identically (97.42% both on the sota holdout); I scope "lossless" strictly as bit-identical holdout accuracy under weight-only int4 RTN.

The myopic ceiling (~88%) is intrinsic: the teacher was privileged with waiting-time state the myopic prompt hides, an Orca-style label-only distillation cost [17] disclosed as a design property, not noise.

6.4 Closed-loop outcome: delay decouples from accuracy

Table 6.3 reports the Qwen3-0.6B closed-loop sweep: fine-tuned (ft) versus identically-compiled stock, three topologies × 30 training-disjoint seeds, paired per-seed against fixed-time.

The honest headline: at n=30 per topology, neither arm significantly beats the fixed-time floor anywhere, and a +42-point offline accuracy gain produces no significant delay change — the **delay-decoupling paradox** anticipated by the decision battery. The one significant effect of fine-tuning is protective: stock control significantly *harms* the congested Bloomsbury grid (+2.6%, p=1.5×10^{-3}), the fine-tuned

Table 6.3: Closed-loop delay vs fixed-time and decision composition (n=30 disjoint seeds per cell; paired t). Negative Δ favours the controller. *auth* = share of all actuations authored by the SLM; *a-s* = share forced by the anti-starvation floor; *acc* = proposal acceptance; *div* = divergence from MaxPressure among SLM-authored decisions. The substitution path accounts for ≤0.5% everywhere and is omitted.

Topology	Arm	Δdelay	p	auth	a-s	acc	div
Euston peak-hour	ft	−3.9%	.16	0.64	0.36	0.70	0.12
	stock	−6.2%	.12	0.62	0.38	0.68	0.47
Bloomsbury grid	ft	+0.7%	.37	0.70	0.30	0.73	0.18
	stock	+2.6%	$1.5{\times}10^{-3}$	0.62	0.38	0.65	0.46
Old Street	ft	−3.1%	.20	0.56	0.44	0.90	0.11
	stock	−1.1%	.35	0.39	0.61	0.54	0.60
Pooled (n=90)	ft	−2.1%	.14				
	stock	−1.6%	.39				
Euston	phi ft	−3.8%	.15	0.64	0.36	0.70	0.13
Bloomsbury	phi ft	+0.6%	.42	0.70	0.30	0.73	0.17
Old Street	phi ft	−1.0%	.56	0.57	0.43	0.88	0.12
Pooled (n=90)	phi ft	−1.4%	.20				

controller is neutral there, and the head-to-head paired difference favours fine-tuning by −1.8% (p=.019). I return to the mechanism — the shield bounds worst-case behaviour for both arms, and external calibration puts the entire LLM-vs-MaxPressure headroom at 1–2% [31] — in §7.2. The Phi-4-mini confirmation arm (final rows of Table 6.3) extends the saturation finding end-to-end: the 3.8B fine-tuned student's closed-loop profile is arm-for-arm indistinguishable from the 0.6B's — Euston −3.8% vs −3.9%, Bloomsbury +0.6% vs +0.7%, the same divergence band (0.11–0.18) and authorship profile — and at the lowest served latency of any arm (median p50 0.94 s, worst cell p99 1.46 s). The dissertation's scale-threshold question is thereby answered in both evaluation regimes: 0.6B suffices.

6.5 Authorship: what fine-tuning actually moves

The structural columns of Table 6.3 carry the dissertation's central finding, and reading them correctly requires keeping the three denominators of §3.3 apart — an error I made in an earlier draft and correct here explicitly, because the distinction changes the interpretation.

First, the fact that frames everything else: **the shield's substitution path almost never fires**. Across every arm and topology it accounts for ≤0.5% of decisions, because both stock and fine-tuned models emit a well-formed, in-menu proposal on essentially every tick. Whatever the safety scaffold contributes here, it is not the correction of malformed output. The real competition for authorship is between the SLM and the anti-starvation floor.

Second, **fine-tuning does raise the SLM's true authorship of actuations**, and most where the stock model was weakest: on Old Street from 0.39 to 0.56, on Bloomsbury from 0.62 to 0.70, on Euston from 0.62 to 0.64. The mechanism is visible in the adjacent column: the anti-starvation floor is forced to intervene far less often (Old Street 0.61 → 0.44), because the distilled model serves neglected approaches before the fairness timer must. Proposal acceptance rises correspondingly (0.54 → 0.90). But the honest headline number is the authored share, not the acceptance rate: even the best fine-tuned arm leaves *30–44% of actuations authored by a hard-coded timer*, which any audit of this deployment must attribute to the scaffold rather than to the model.

Third, and cutting against the simple reading, **the distilled policy is markedly more MaxPressure-like**: divergence among SLM-authored decisions falls from 0.46–0.60 to 0.11–0.18, so the fine-tuned model chooses what MaxPressure would have chosen on roughly nine decisions in ten. This admits a deflationary

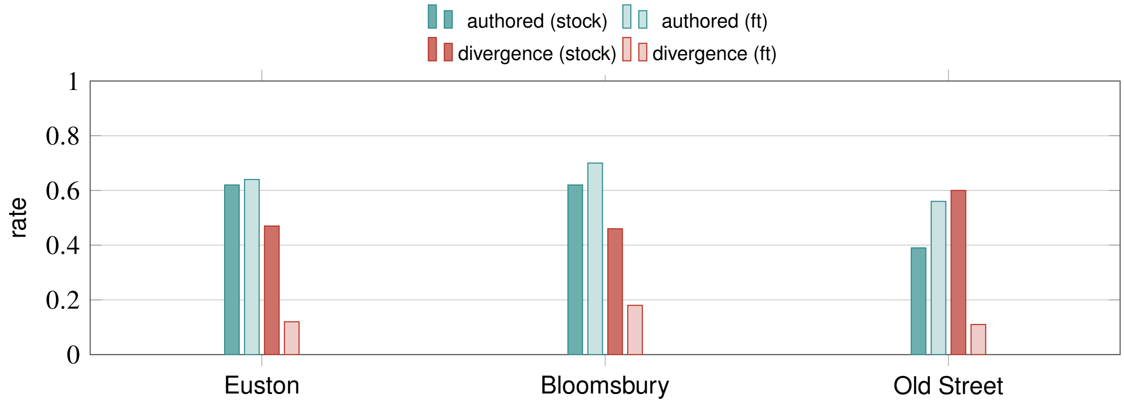

Figure 6.1: What fine-tuning moves (n=30 seeds per bar). Teal: the SLM's share of *all* served actuations rises, most sharply on Old Street (0.39 → 0.56); the remainder is authored by the anti-starvation floor. Red: divergence from MaxPressure among SLM-authored decisions falls roughly four-fold — the distilled policy coincides with the classical reference far more often.

rival explanation — that distillation largely taught the student to imitate the classical reference — and the evidence neither establishes nor excludes it. Three observations bound it. The teacher was a frontier model reasoning over accumulated waiting, not MaxPressure, and the students agree with the teacher's held-out labels at 97–99% (§6.3), so the convergence is at most indirect. The fine-tuned arms do not reproduce MaxPressure's closed-loop behaviour — MaxPressure is significantly worse than fixed-time on these topologies (§6.1) while the fine-tuned arms are not. And the remaining one-in-ten divergences are where the arms differ from both baselines. Distinguishing imitation from independent rediscovery of a similar policy would require a decision-level counterfactual study I did not run, and I flag it as the most substantive open question about this result.

What survives all three qualifications is the audit-relevant claim, stated at its true magnitude. An accident audit over the stock Old Street deployment could attribute only 39% of actuations to the learned component; over the fine-tuned deployment it attributes 56%, with the balance falling to a deterministic timer whose behaviour is trivially explicable. Fine-tuning buys a controller whose served decisions are substantially its own and which needs the fairness scaffold less — a smaller claim than *the model is the controller*, and one the data supports.

6.6 Failure forensics: the audit layer explains the tail

The audit architecture's value shows most clearly when control fails. Across all 270 closed-loop cells, 14 (5.2%) are catastrophic (delay ≥ +20% vs baseline) — and every one of their signed decision chains verifies, so each failure has a provable record to interrogate. Three findings emerge from that interrogation.

First, **failures are a controller property, not seed hardness**: the catastrophic seeds are baseline-easy or median demand realisations (baseline delay ranks 1–20 of ~35, never the hard tail), and no seed breaks all three arms.

Second, the two failure regimes have different, attributable mechanisms. The stock model's five failures are its own served decisions: in the worst cell (Euston s13, +110.3%), the verified chain shows 214 of 335 decisions SLM-authored with 119 divergent proposals served (override 0.556), zero shield fallbacks, and anti-starvation at its normal rate — the scaffold did not cause the collapse, the model's policy did, and the chain proves it. The same-seed fine-tuned arms sit at +1.1% and +5.5%.

Third, **the two fine-tuned students converge at the trajectory level on a substantial minority of cells**. Comparing the 0.6B and 3.8B students seed-for-seed across the 90 shared cells: on nine they produce *bit-identical* mean delay, on a further 25 they agree within 0.5 points, and on the remaining 56 they differ. The identical cells are not duplicated records — wall-clock times differ by up to a factor of two and median latencies by 2.1× — but genuinely separate runs that served the same phase sequence; in one such cell both arms took 307 decisions and their SLM/floor attribution differs by exactly one tick, where the floor and the model happened to select the same phase. The catastrophic cells fall inside this converged subset, which is why the two students fail together.

Two independently trained students, different in architecture and 6× apart in scale, therefore reproduce each other's control trajectory on roughly a third of cells and diverge on the rest. I read this as the behavioural counterpart of the offline saturation result — the distillation target appears to constrain not only what the students score but what they *do* — while stating the limit of the evidence: partial trajectory convergence is equally consistent with both students defaulting to similar behaviour on the easier cells, and I have not separated those explanations.

One honest gap: this campaign persisted aggregate decision statistics per cell, not per-step audit records, so *when* within a run the divergence concentrated is unrecoverable here; the per-step chain is retained in the dedicated accident-suite runs (§6.7). Full tables in `results/failure_forensics.md`.

6.7 Audit and crash reconstruction

The audit layer was evaluated on seven pre-defined collision scenarios spanning the legally distinct cases (red-light running, EV exemption, maintenance vehicle, amber gambling, conflicting green, dark signal, spoofed EV). All seven prove end-to-end: hash-chain verification passes, signature verification passes, and an imposter re-signing attack is caught in every scenario. The governing UK rule is *inferred deterministically from the verified facts* — e.g. the spoofed-EV scenario correctly denies the exemption and anchors fault on the driver, while the conflicting-green scenario is the sole case attributing (weak, misfeasance-only) authority fault — and the system emits a cited evidence pack for a human adjudicator, never a binding verdict, by design [3].

6.7.1 A worked evidence pack

Figure 6.2 traces the hardest of the seven scenarios end to end, because it is the one where the *system itself* may be at fault: a collision at a junction that emitted a conflicting green.

Three properties are worth drawing out. First, the pack is built only from facts the chain verifies — `signal_state`, the `conflicting_green` flag, the crossing class — and each cited rule carries the specific fact that triggered it, so an investigator can audit the inference, not merely read a conclusion. Second, the legal reasoning is faithful to a doctrine that is unfavourable to the deployer: under *Gorringe v Calderdale* a highway authority's mere omission attracts no duty, so authority fault defaults to zero, and it becomes non-zero here only because *Bird v Pearce* treats a positively-wrong indication as misfeasance — fault weight `low`, flagged `common_law` rather than `MUST`, with legal causation explicitly `NOT_ASSESSED`. The system does not overstate a case against a driver when its own record shows the signal misled them, and it does not overstate the case against itself either. Third, the tamper probe passes: an imposter re-signing the incident record is rejected on signature verification even though the chain hashes are recomputed consistently — the property that makes the pack evidence rather than assertion.

6.8 Trust-coefficient battery

The trust mechanism passes its full property battery (9/9): a persistently honest junction climbs to 0.929 trust; a persistent liar collapses to 0.011 and loses the power to corroborate; the asymmetry is deliberate (ten verified truths to climb from 0.5 to 0.9, one lie from high trust falls 0.65); and a truthful-but-late report is vindicated retroactively by local sensing, so honesty is never permanently penalised for latency. Trust is discount-only by construction — it can strip a liar's corroboration power but can never add a preemption on zero evidence, and local sensing is never itself doubted.

In closed-loop traffic (honest / blatant / stealth misreporting injected into SUMO), I ran the battery against *both* stock and fine-tuned controllers at both scales — the fine-tuned deployment is the one that matters,

```
VERIFIED RECORD (seq 0, junction J, hash a0bb851c..., t=100.0)
  chain verified: True   signatures verified: True
  imposter re-sign attempt: CAUGHT
VERIFIED FACTS
  signal_state = green      conflicting_green = True
  crossing_class = civilian second_party_on_green = False
  key_present = False       corroborated = False
GOVERNING RULES (inferred from the facts above)
  LR-authority-misfeasance [Bird v Pearce]  fault: low  (common law)
    "the audit shows the system emitted a conflicting green
     (a positively-wrong indication) -- misfeasance, not omission"
  LR-authority-nonfeasance [Gorringe v Calderdale] fault: low (common law)
    "authority fault otherwise defaults to zero; it turns non-zero only
     on this misfeasance/conflicting-green branch (low confidence;
     legal_causation NOT_ASSESSED)"
OUTPUT: cited evidence pack for a human adjudicator -- not an adjudication
```

Figure 6.2: The conflicting-green evidence pack, verbatim from `results/experiment_crash_law.json`: the only one of the seven scenarios in which authority fault is non-zero, and the case where the system's own record is used against its operator.

precisely because of §6.5: a controller that actually acts on its coordination inputs is the one poisoned inputs can move. Three findings (three seeds per cell, descriptive).

First, **detection is controller-independent by design and in measurement**: the ledger scores claims against local sensing, and under every controller a blatant liar is caught and collapsed (Bloomsbury: 62–70 catches, final trust 0.21–0.27; grid4×4: 146–151 catches, trust ≤0.03, corroboration power stripped), while a calibrated stealth attacker evades essentially everywhere (≤2 catches, trust ≥0.92) — the stealth evasion stands as an open limitation. On the corridor and Old Street the liar junction is never consulted, so the mechanism is inert there (zero lies injectable) — reported, not hidden.

Second, **the delay cost of blatant lying survives detection**: on the synthetic grids blatant misreporting adds roughly +3 to +11 points of delay relative to the same controller's honest run even though the liar's trust collapses — trust bounds the liar's future influence but does not undo congestion already caused. On Bloomsbury, the real grid, local-sensing override holds the damage to within ~3 points (fine-tuned qwen: −1.3% under blatant attack vs −1.1% honest).

Third, **stealth's harm is positional, not congestive**: the stealth attacker's small, intermittent inflations cost approximately nothing in delay (−2 to +3 points vs honest) — what stealth buys is retained corroboration power at high trust, a position it could later abuse. The threat model that matters for a fine-tuned deployment is therefore asymmetric: blatant attacks are caught but briefly expensive on congested grids; stealth attacks are cheap, patient, and undetected. Both conclusions argue that the trust ledger is a necessary companion to fine-tuning rather than an optional extra, and that detection calibrated to distributional implausibility (§7.4) is the right next defence.

6.9 The fair re-test of the SLM authorisation negative

An early measured negative — the raw SLM losing to a deterministic template on its legal-note job — carried a supervisor-posed objection (Stott): the model was guessing because the policy was never in front of it, so the negative cannot stand until the SLM is re-tested with explicit in-prompt rules and a reason-then-classify structure. I ran that re-test on the balanced 26-claim authorisation dataset of §6.7's suite, four models × two arms (policy-scaffolded vs bare), scored against the deterministic knowledge-base gate.

The objection is partially vindicated: for a capable stock model the scaffold works. Stock Phi-4-mini rises from 42.3% to 69.2% class accuracy, its false-negative rate falls to zero, and it stops denying any genuine emergency. But the negative itself is robust: even the best cell — stock Phi, scaffolded — still makes **four dangerous false grants** (granting preemption to claims that must be denied, FPR 0.33) where the deterministic gate makes none in 26/26. Policy grounding rescues recall, not safety.

The re-test also produced this dissertation's clearest evidence of specialisation cost: the *fine-tuned* students are far worse here than their own stock bases — ft-Phi falls to 19.2% scaffolded with 15 of 26 responses unparseable, and ft-Qwen cannot emit the required `{"class", "grant"}` schema at all bare (26/26 parse failures), because distillation captured its output format onto `{"phase": N}`. The retention probe's mild MMLU decline (§7.3) understates the cost on near-neighbour structured tasks. Both conclusions reinforce the architecture: the deterministic gate owns authorisation, the fine-tuned model owns phase selection, and neither should borrow the other's job.

6.10 Serving reliability at the edge

Sustained on-device serving produced four documented reliability findings across the campaign: (i) the catalogue Qwen3-0.6B GPU artifact emitting corrupted tokens after long service uptime; (ii, iii) two unprompted Foundry Local service self-restarts mid-evaluation, killing runs at 5,500/5,580 and 4,300/5,580 rows; (iv) registration of builder-compiled models requiring undocumented synthesis of the inference manifest. Each was mitigated in harness — endpoint rediscovery, model reload with retry-of-failed-rows and statistics intact, fail-loud parse accounting that aborts rather than scores error text — and the mitigations are themselves contributions: they are what turned a research prototype into a harness that survived a six-week unattended campaign.

Decision latency, measured per cell across the closed-loop sweeps, is summarised in Table 6.4. The 3.8B student is the fastest arm served (median p50 0.94 s) and the 0.6B students the slowest (1.74–1.75 s) — a counter-intuitive ordering that is a property of the compiled artifacts and their execution paths rather than of parameter count, and one I report because it cuts against the assumption that the smaller model is always the cheaper one to serve. Two individual calls in the 0.6B arms exceeded the 10 s decision interval (worst 13.8 s), so the real-time margin is comfortable at the median and not guaranteed in the tail: on those ticks the junction acted on stale state. This is the same criterion that excludes phi-4-mini-reasoning (§5.4); the difference is frequency — isolated outliers here against a p99 above the interval there — and a deployment would need a hard timeout with deterministic fallback, which the architecture already supports.

Table 6.4: Decision latency across closed-loop cells (per-cell summaries, GPU-served, sota arms). The decision interval is 10 s.

Arm	p50 (median)	p50 range	p99 (worst cell)	worst call
Qwen3-0.6B ft	1.75 s	1.39–2.57 s	7.58 s	13.79 s
Qwen3-0.6B stock	1.74 s	1.46–2.51 s	2.97 s	9.88 s
Phi-4-mini ft	0.94 s	0.92–1.20 s	1.46 s	1.49 s

Chapter 7

Discussion

7.1 Answering the research questions

RQ1 (edge competence): qualified, and the qualification is topology-dependent. At powered replication (n=30 disjoint seeds) the picture splits. On Old Street, stock SLM control significantly beats the fixed-time floor (up to -11.8%, p<0.001) and one arm also beats MaxPressure. On the measured-demand Euston corridor the same controllers are 7–20% *worse* than the floor, significantly so on half the arms — a cell my exploratory three-seed sweep had reported as a win, and the clearest evidence in this work for why powered replication on measured demand was necessary. Fine-tuned control removes that corridor deficit without becoming significantly better than the floor anywhere. The defensible claim is therefore narrow: on-device SLM control is competent within the deployable envelope on complex junction geometry, harmful on a saturated arterial before distillation, and never demonstrably superior to a naive fixed-time plan at n=30.

RQ2 (what distillation buys): authorship, and it saturates at 0.6B. Offline, distillation is dramatic (55–69% → 97–99%) and scale-flat — a 3.8B student adds nothing over 0.6B, and no format specialist beats the generalist even on its own format. Closed-loop, the accuracy does not become delay; it becomes *ownership*: the SLM's share of served actuations rises from 0.39–0.62 to 0.56–0.70, divergence from MaxPressure among its own decisions falls four-fold, and the stock harms (congested grid +2.6%, p=1.5×10^{-3}; measured-demand corridor +7 to +20%) are repaired to neutral. For a shielded, audited controller I argue this is the correct success criterion, and §7.2 explains why delay was never the sensitive variable.

Two further results qualify the answer in opposite directions. The saturation is deeper than the accuracy tables alone show: the two students fail on identical seeds with matched magnitudes (§6.6), so distillation converges different architectures onto one effective policy — which also means a second, larger student buys no diversity, and an ensemble of the two would be close to worthless. Against that, distillation is not free: on an adjacent structured task the fine-tuned students are worse than their own stock bases, one of them unable to emit the required output schema at all (§6.9). The capability is not so much lost as *narrowed onto the deployed contract*, which is acceptable — even desirable — for a component whose job is one decision type behind a deterministic gate, and unacceptable if the same weights were expected to serve the authorisation or legal-note roles as well.

RQ3 (provable audit): yes, as designed. Seven scenario reconstructions prove end-to-end with tampering caught; the system emits cited evidence packs, and the legally weak branches (authority fault) are weak *in the law*, faithfully represented rather than overstated.

RQ4 (trust): partial, honestly — and the partiality is structured. The mechanism's properties all hold; blatant misreporting is detected and the liar's corroboration power stripped under every controller tested, stock and fine-tuned alike, because the ledger scores claims against local sensing rather than against model behaviour. Three limits qualify that. Detection does not undo damage already done (a caught blatant attack still costs up to eleven delay points on congested grids); a calibrated stealth attacker is not detected at all, and its payoff is positional — retained trust it could later spend — rather than congestive; and the mechanism is inert where coordination traffic is minimal, as on the corridor. The finding that most changes the design picture is that the delay cost of a blatant attack is *larger* against fine-tuned controllers than stock ones: authorship cuts both ways, because a model that genuinely acts on its inputs is moved further by poisoned inputs. A trust ledger is therefore a necessary companion to distillation rather than an optional layer, and it is not a sufficient defence.

7.2 Why accuracy decouples from delay

Three mechanisms jointly explain the paradox, and all three are visible in the data rather than conjectured. First, *the shield compresses the outcome distribution*: masking, substitution, and anti-starvation bound the worst case for every arm, so the behavioural difference between a 59% and a 97% proposer is far smaller served than raw. Second, *the headroom was small*: external calibration from a 13× larger tuned model puts the entire LLM-vs-MaxPressure delay gap at 1–2% [31], and my own decision battery showed a near-chance proposer winning closed-loop cells — per-decision agreement with a delay-optimal heuristic is simply not the trajectory-level objective. Third, *where the stock model could do damage, fine-tuning shows up*: the single significant closed-loop effect is the repair of stock harm under grid congestion, exactly where bad proposals compound fastest. The practical reading for deployment: fine-tune to own the decisions and to remove harm tails, and do not expect — or promise — delay headlines inside a safety shield.

7.3 Threats to validity

I declare four threats with their defences, and none is cosmetic.

Label-only distillation (construct). I distil teacher labels, not reasoning traces; Orca-style trace distillation [17] might transfer more competence. The myopic format makes the cost measurable: its ~88% ceiling is the privileged-teacher gap, disclosed as a design property.

Retention (internal). Aggressive SFT can erode general capability with scale-dependent forgetting [12]. I measured it: a seeded 200-item MMLU sample served through the same endpoints shows the 3.8B student dropping from 57.5% (stock) to 53.0% (fine-tuned) — a decline consistent with mild forgetting but not significant at this sample size (z=0.91, p=.37) — while retaining its answer-format discipline (94% strict JSON). At 0.6B the probe is uninformative by floor effect: the stock model already scores at chance (26.0%) on this format, so there was no measurable general capability to lose. The honest conclusion is that no significant retention damage was detected on general knowledge, with the caveat that a 200-item probe bounds only large effects. That conclusion should not be read as reassurance, because a nearer-neighbour probe tells a harsher story: on the authorisation task (§6.9) both fine-tuned students fall well below their own stock bases, one failing to produce the required output schema at all. The damage is not diffuse forgetting of world knowledge but narrowing onto the trained contract — which general-knowledge benchmarks are poorly placed to detect, and which matters more for a deployment that might later ask the same weights to do a second job.

Quantisation scope (construct). "Lossless" int4 is claimed strictly as bit-identical holdout accuracy under weight-only RTN — not as a general equivalence claim, which the quantisation literature would not support [5, 24].

Simulation and demand realism (external). Everything is SUMO; real-map demand magnitudes are measured (DfT hourly profile on Euston) but their vehicle-level realisations are synthetic draws, and comparators exclude trained deep-RL systems. The MaxPressure comparison is accordingly reported without a general claim: the three-seed reversal did not replicate at thirty paired seeds (§6.1), six grid studies showing MaxPressure's advantage are cited, and my own grids reproduce their direction. Legal scope: the audit scenarios use synthetic actors on real topology, so no real data subject exists; a real deployment would face the adjudication and disclosure exposures analysed in the project's legal grounding, which curation itself is an adjudicative act (*Bates v Post Office (No 6)*) and qualified privilege is defeasible.

7.4 Limitations and future work

Four directions follow directly from the findings. **Formal shield synthesis**: replace the heuristic composite with a synthesized shield from a temporal-logic safety specification [1] — the per-junction product game is small enough for exact synthesis, and the authorship metrics defined here would then carry minimal-interference guarantees. **Closed-loop objectives**: GRPO-style tuning against trajectory reward [31] attacks the decoupling directly; label distillation is the wrong objective if delay is the goal. **Stealth-robust trust**: detection calibrated to distributional implausibility rather than single-report bounds. **Hardware-in-the-loop**: the serving-reliability findings (service self-restarts, artifact corruption) are exactly the failure class a cabinet deployment must survive, and they argue for a watchdog-supervised serving architecture before any field pilot.

Chapter 8

Conclusion

This dissertation asked whether an on-device small language model, served by Microsoft Foundry Local within a 6 GB envelope, can control real London junctions competently while producing an accident audit worth trusting — and at what scale the learned component earns its place.

The answer to feasibility is a qualified yes, and the qualifications are the substance. Shielded on-device SLM control significantly beats the fixed-time floor on the study's most complex junction geometry, is harmful on a saturated arterial before distillation and neutral after it, and is never demonstrably better than a naive fixed-time plan at thirty seeds. The classical baseline it is measured against turns out to be unreliable rather than uniformly weak on real geometry — a three-seed result I retracted on replication. The full control loop, from frontier-teacher distillation through QLoRA training to int4 serving, fits on one consumer device; the 0.6B student trained on the same GPU that serves it.

The answer to scale is sharper than expected: distillation saturates at 0.6B. A student six times larger lands on statistically identical decision accuracy, no format specialist beats a generalist even on its own format, and a student trained without one topology loses nothing on it. Within this deployable envelope, capability is not the bottleneck — the distillation ceiling is.

The central finding reframes what fine-tuning is *for* in a shielded controller. Closed-loop, at thirty disjoint seeds per topology, the 97%-accurate student does not significantly beat the floor — and neither does anything else inside the shield's envelope, whose compression of outcomes, together with the small delay headroom the wider literature itself reports, explains the decoupling. What fine-tuning demonstrably moves is authorship: the model's share of served actuations rises from 0.39–0.62 to 0.56–0.70 as the fairness floor is needed less, its divergence from MaxPressure falls roughly four-fold, and the stock model's measured harms — a significantly worse congested grid, and a 7–20% deficit on the measured-demand corridor — disappear. Because the audit layer proves who decided what — seven collision scenarios reconstructed end-to-end, tampering caught, governing UK law inferred from verified facts into cited evidence packs — authorship is precisely what makes the audit about the certified model rather than its fallback. Fine-tuning buys accountability, not delay.

The dissertation's credibility rests as much on its nulls as its wins: the corridor null at n=30, the stealth attacker that evades the trust ledger, the privileged-teacher ceiling on the myopic format, and the four serving-reliability failures are reported with mechanisms, because a feasibility study that hides its boundary conditions has not established feasibility. Within those boundaries, the case is made: an audited, shielded, on-device SLM junction controller is possible today, at 0.6B, on hardware a signal cabinet could house — and the path from here runs through closed-loop objectives, synthesized shields, and a watchdog-supervised field pilot.

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Chapter A

Reproducibility artifacts

A.1 Environment and version pins

Every measurement in this dissertation was produced under the pins in Table A.1. The Foundry Local version is pinned deliberately: custom-model serving behaviour (registration format, execution-provider options) is version-sensitive, and an upgrade mid-campaign would have invalidated cross-run comparability. One consequence is recorded honestly in §6.10: a newer model generation (Qwen3.5) ships a configuration schema this runtime cannot parse, so it is reported as blocked-by-runtime with the exact error rather than as a capability result.

Table A.1: Version and hardware pins.

Component	Pin
Serving runtime	Microsoft Foundry Local 0.8.119
Compile path	onnxruntime-genai model builder, int4 RTN, weight-only
Execution providers	CPU and WebGPU (decision-equivalent, §4.2)
Simulator	SUMO with TraCI, 1,200-step horizon, ≈10 s decision interval
Serving/measurement GPU	NVIDIA RTX 2060, 6 GB (the deployment envelope)
Training GPU (3.8B)	UCL departmental NVIDIA RTX 4090, 24 GB
Training GPU (0.6B)	the RTX 2060 itself (13.5 h)
Host OS	Windows 10 (build 19045) — predates Windows ML EP autoregistration

A.2 Seeds and dataset splits

Training-state seeds: $\{3, 7, 11, 29\}$. Junction states for distillation were sampled only from these. **Closed-loop evaluation seeds:** $\{1..30\} \setminus \{3, 7, 11, 29\} \cup \{31, 32, 33, 34\}$ — thirty seeds, disjoint from training (§5.3). **Offline holdout:** split by deterministic content hash (`HOLDOUT_MOD=12`) and frozen before the first training run; 5,580 held-out examples against 59,533 training examples, from 65,113 accepted renderings (6 rejected by the sanity verifier for serving an empty phase without a justifying note). Per format: sota 16,982, myopic 16,982, prediction 18,596, coordination 12,553.

A.3 Prompt contract

All four formats share one output contract — exactly `{"phase": N}` — and the *same rendering code serves training, offline evaluation, and closed-loop control*, so no format skew can flatter the offline numbers. The delay-aware (sota) system prompt is reproduced verbatim:

```
You control ONE traffic-signal junction (this junction only; no neighbour information).
For each green phase you are given: the number of waiting (queued) vehicles,
the TOTAL accumulated waiting time of those vehicles in seconds, and whether it
is the phase currently green. Choose the phase to serve next to MINIMISE total
vehicle waiting time and unnecessary stops - not simply the longest queue.
```

The myopic format withholds accumulated waiting; prediction adds short-horizon arrival counts; coordination adds neighbour release notes. Parsing falls through four tiers (strict JSON, braced, keyed, bare integer) with the tier recorded per decision.

A.4 The compile-and-serve chain

Reproducing a served custom model requires the following steps in order; each was established by failure during this project and none is documented by the vendor toolchain.

1. Train QLoRA adapters (r=16, NF4 double quantisation, fp16 compute, completion-only loss, family-correct chat template).
2. Merge adapters to a bf16 checkpoint.
3. Compile with the onnxruntime-genai model builder (`-p int4 -e cpu`) using the pinned transformers pairing — newer environments fail on attention-module or loss-utility imports.
4. Patch `tokenizer_config.json`: `tokenizer_class` must be a value the runtime accepts (`GPT2Tokenizer`); transformers-5 era configs emit a backend name that fails to load.
5. For Phi checkpoints, strip remote-code `auto_map` entries.
6. **Synthesise** `inference_model.json` (model name plus prompt template) — the builder emits none, and without it the model registers but returns HTTP 400 on every request.
7. Copy into the Foundry Local cache directory and, for GPU serving, set `provider_options` to WebGPU in `genai_config.json`.

A.5 Harness invariants

Three properties of the experiment harness underwrite the results. *Fail-loud:* API errors are counted and abort a run rather than being scored as answers; an unservable model records a skip, never a score. *Resumable:* cells are keyed by (model, config, topology, seed) and skipped if already measured, so a crash loses at most one cell — one zero-byte cell caused by a mid-write power loss was detected by JSON validation and re-run. *Recoverable:* after two observed mid-run service self-restarts, the offline evaluator rediscovers the endpoint, reloads the model, and retries failed rows with statistics intact.

Chapter B

Extended results

B.1 Per-topology closed-loop detail

Chapter 6 reports means and paired tests; the per-seed cells behind them are retained as JSON in `results/frontier_raw/`, one file per (model, config, topology, seed) containing the baseline delay, controller delay, relative change, decision composition (SLM / substitution / anti-starvation), from which authored share, proposal acceptance and divergence rate are derived, plus the latency distribution and audit-chain verification result. Every figure in this dissertation is recomputable from those files with the aggregation scripts in `src/`.

B.2 Failure forensics tables

The catastrophic-cell analysis of §6.6 — 14 of 270 cells at ≥ +20% delay, their seed-matched comparators, and the shared-versus-arm-specific decomposition — is tabulated in `results/failure_forensics.md` with the underlying data in `results/failure_forensics.json`.

B.3 Adversarial trust battery

Per-topology, per-scenario cells for all three controllers (stock and fine-tuned, both scales) are in `results/experiment_trust_traffic_<model>.json`, with raw cells under `results/trust_traffic_raw/` keyed by model, topology, scenario, and seed.

B.4 Authorisation re-test

The four-model, two-arm fair re-test of §6.9 is recorded in `results/slm_auth_lee_<model>.json`, each containing class accuracy, grant accuracy, precision, recall, false-positive and false-negative rates, dangerous false grants, parse failures, and P50/P95/P99 latency per arm.

Chapter C

UK-law fault-grounding knowledge base

The reconstruction layer matches verified collision facts against a curated knowledge base of UK traffic law. The base is deliberately small, explicit, and citable: each rule carries a statute or case reference, a binding strength (`MUST` for statutory prohibitions, `common_law` for judicial doctrine), a fault weight, and the verified fact that triggers it. Table C.1 lists the rules exercised by the seven evaluated scenarios.

Table C.1: Fault-grounding rules exercised by the evaluated scenarios.

Rule	Authority	Effect
Driver crossed red	RTA 1988 s36	Criminal offence; driver fault high
Red prohibition	TSRGD 2016 Sch14	Must not proceed beyond stop line
Amber	TSRGD 2016 5(9)	Driver fault high where a safe stop was possible
EV exemption	TSRGD / conditional	Conditional on authorisation *and* corroboration; endangerment test still applies
Maintenance vehicle	no exemption	Works vehicle treated as an ordinary driver
Dark signal	Highway Code r.176	Duty to obey falls away; ordinary care governs
Authority misfeasance	*Bird v Pearce*	Positively-wrong indication: authority fault non-zero but low
Authority nonfeasance	*Gorringe v Calderdale*	Mere omission: authority fault defaults to zero

Two properties of this design matter for the accountability claim. First, the doctrine is *unfavourable to the deployer of this system*: fault attaches to the authority only through the misfeasance branch, which is exactly the branch the audit chain can prove or disprove from its own records, and the system reports it against itself when the facts support it (§6.7.1). Second, a spoofed emergency-vehicle claim receives no exemption — the conditional exemption requires a valid authorising key and corroboration, so a forged claim leaves the driver squarely within the red-light prohibition. Legal causation is never assessed by the system; it is marked `NOT_ASSESSED` and left to a human adjudicator, in line with the treatment of legal hallucination risk in §2.6.